



DRDO Research Internship

Project Report

Analyzing the Effect of Background Change on Vision Model Predictions

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Organization: DRDO – SAG

Technical Stack:

PyTorch · CLIP (ViT-B/32) · ResNet-18 · OpenCV · rembg · Cosine Similarity

Dataset:

LAION-400M (Streaming Subset)

Internship Duration: 4 weeks

Abstract

Modern vision models often rely on both foreground objects and surrounding background context to interpret images. This project investigates how changes in image backgrounds affect the semantic predictions of a vision-language model.

A three-stage pipeline was implemented: (1) automated foreground extraction and background separation, (2) background feature embedding and similarity analysis using cosine similarity, and (3) foreground compositing onto visually dissimilar backgrounds, followed by semantic evaluation with the CLIP (ViT-B/32) model.

Results indicate that background context can significantly influence model confidence and predicted labels. Foreground objects such as humans tend to dominate predictions, but object-centric scenes (e.g., vehicles, animals, beds) exhibit measurable shifts in semantic interpretation after background replacement. Limitations in compositing realism introduced visual artifacts, which may have constrained stronger conclusions.

These findings highlight the importance of considering background context in vision models and emphasize the need for realistic image synthesis in evaluating model behavior.

Keywords: Vision Models, CLIP, Contextual Bias, Background Replacement, Semantic Evaluation, LAION-400M

Table of Contents

Section 1: Introduction

- 1.1 Background
- 1.2 Motivation
- 1.3 Objective
- 1.4 Project Workflow

Section 2: Dataset and Data Preparation (Week 1)

- 2.1 Dataset Selection
- 2.2 Foreground Extraction (rembg)
- 2.3 Dataset Filtering
- 2.4 Outputs Generated

Section 3: Background Similarity Analysis (Week 2)

- 3.1 Background Feature Representation (ResNet-18 Embeddings)
- 3.2 Cosine Similarity Computation
- 3.3 Identification of Dissimilar Backgrounds

Section 4: Background Replacement and Semantic Evaluation (Week 3)

- 4.1 Background Replacement Pipeline
- 4.2 Image Compositing Techniques
- 4.3 Limitations of Compositing
- 4.4 Semantic Evaluation using CLIP (ViT-B/32)
- 4.5 Label Set and Confidence Analysis

Section 5: Results and Analysis

- 5.1 Dominant Label Distribution
- 5.2 Average Confidence per Label
- 5.3 Background Sensitivity Analysis
- 5.4 Interesting Findings

Section 6: Limitations

Section 7: Key Learnings

Section 8: Conclusion

Section 9: Future Work

References

1. Introduction

1.1 Background

Deep learning models for computer vision have achieved remarkable accuracy in a variety of tasks, including object detection, image classification, and scene understanding. Despite these successes, a key question remains: do these models truly “understand” the objects in an image, or do they rely heavily on contextual cues from the surrounding background?

Humans tend to focus primarily on the foreground object while using background information only as supporting context. In contrast, deep neural networks may learn strong correlations between objects and their typical environments. For instance, cars are often associated with roads or urban settings, while beds are commonly seen in indoor scenes. Such correlations may lead models to infer the presence of objects not solely based on visual features, but also on background context.

This observation motivates the central question of this project: **to what extent do vision models depend on background context for their predictions, and how does altering the background influence their semantic understanding of an image?**

1.2 Motivation

Most modern vision models are trained on large-scale image datasets, many of which contain implicit statistical biases linking objects to certain environments. While these biases can improve performance on standard benchmarks, they may also cause models to rely on background patterns rather than intrinsic object features.

Understanding how context influences model predictions is critical for several reasons:

1. **Interpreting model predictions:** Identifying whether predictions are truly object-driven or context-dependent helps explain model behavior.
2. **Improving robustness:** Models that are overly sensitive to background context may fail in unfamiliar or manipulated environments.
3. **Detecting context-based bias:** Awareness of context reliance is essential for designing fair and reliable computer vision systems, particularly in safety-critical applications.

By investigating the role of background context, this project aims to shed light on these important aspects of vision model behavior.

1.3 Objective

The primary objective of this project is:

To analyze how changes in image background influence the semantic predictions of a vision-language model, specifically CLIP (ViT-B/32).

This involves quantifying the effect of background replacement on model confidence, label prediction, and overall sensitivity to context.

1.4 Project Workflow

The project was structured into three main phases, corresponding to the three weeks of the internship:

Week 1 — Data Preparation

- **Foreground extraction:** The primary objects in the images were segmented from the original backgrounds using the **rembg** library based on the U2-Net model.
- **Background separation:** Background images were isolated to allow independent analysis and manipulation.
- **Mask generation:** Binary masks were created to support accurate compositing of foregrounds onto new backgrounds.

Week 2 — Background Similarity Analysis

- **Feature extraction:** Each background image was converted into a feature vector using a pretrained ResNet-18 model, capturing high-level visual characteristics.
- **Similarity computation:** Cosine similarity was applied to measure visual similarity between backgrounds, enabling the identification of dissimilar scenes suitable for replacement experiments.

Week 3 — Background Replacement and Evaluation

- **Foreground compositing:** Foreground objects were placed onto visually dissimilar backgrounds using alpha blending, while attempting to preserve realism.
- **Semantic evaluation:** The modified images were evaluated using the CLIP (ViT-B/32) model to analyze changes in predicted labels and confidence scores caused by

background alterations.

This structured workflow allowed systematic analysis of the impact of background context on vision model predictions, from data preparation to experimental evaluation.

2. Dataset and Data Preparation(Week 1)

2.1 Dataset Selection

For this project, the **LAION-400M** dataset was chosen as the primary source of images. LAION-400M is a large-scale, open-source dataset consisting of hundreds of millions of image-text pairs collected from the internet. Unlike curated datasets such as COCO, which have editorial selection bias, LAION provides diverse, real-world imagery that better represents natural visual contexts and varied object-background correlations.

To optimize storage and processing, the images were **streamed directly from their URLs** and processed dynamically, without downloading the entire dataset. This approach allowed efficient handling of large-scale data while maintaining flexibility for experimentation.

2.2 Foreground Extraction

The first step in preparing the dataset was the **automatic extraction of foreground objects** from each image. This process was performed using **rembg**, which is based on the **U²-Net segmentation model**. U²-Net is a state-of-the-art architecture for salient object detection, capable of accurately separating the main object from its background.

Processing Steps

1. Each input image was converted to **RGB format**.
2. **rembg** was applied to generate an **alpha segmentation mask**.
3. A **binary mask** was extracted from the alpha channel to separate foreground and background.
4. The mask was refined using **morphological operations** to remove noise and improve boundary accuracy.
5. **Foreground and background images** were generated for further experimentation.

Outputs Generated

For each processed image, the pipeline produced the following:

- **Original Image** – the unaltered image from the dataset.
- **Binary Mask** – a refined mask highlighting the foreground object.

- **Foreground Object** – the extracted main object with transparent background.
- **Background Image** – the isolated background without the foreground object.

Original



Mask



Foreground



Background



Figure 2.1: Example of original image, mask, foreground object, and background.

2.3 Dataset Filtering

To ensure the quality and consistency of the dataset, several filtering criteria were applied:

- **Resolution:** Images with too low resolution were removed.
- **Foreground Area:** Images where the extracted foreground occupied an extremely small portion of the frame were discarded.
- **Mask Quality:** Images with poorly segmented masks were excluded to prevent errors in later stages.

After applying these filters, **8,727 high-quality images** remained and were used for subsequent experiments, ensuring both diversity and reliability in the evaluation pipeline.

3. Background Similarity Analysis (Week 2)

Understanding the visual characteristics of image backgrounds is essential for analyzing how background changes influence vision model predictions. This section describes the process of representing background images as feature embeddings, computing similarity scores, and identifying dissimilar backgrounds for replacement experiments.

3.1 Background Feature Representation (ResNet-18 Embeddings)

To quantitatively analyze background content, each background image was converted into a **feature embedding** using a **pretrained ResNet-18 convolutional neural network**. ResNet-18 is a widely used architecture that efficiently extracts high-level visual features, capturing textures, patterns, and scene layouts.

Importantly, **only the background regions** of images were used to compute embeddings. This ensures that the extracted features reflect the environment exclusively, without interference from foreground objects, which could bias the analysis.

Processing Steps

1. **Preprocessing:** Each background image was resized to 224×224 pixels, the input size expected by ResNet-18.
2. **Normalization:** Pixel values were normalized using the standard ImageNet mean and standard deviation to match the pretrained model's input expectations.
3. **Feature Extraction:** The background image was passed through ResNet-18, and the **activations from the final layer before classification** were extracted as the feature embedding.
4. **Embedding Storage:** All embeddings were stored for subsequent similarity computations.

Using a pretrained model ensures that embeddings capture general visual patterns learned from millions of real-world images, without requiring additional training.

3.2 Cosine Similarity Computation

To measure visual similarity between backgrounds, **cosine similarity** was applied to the feature embeddings. Cosine similarity compares the **angle between two vectors** in high-dimensional space, focusing on their directional alignment rather than magnitude differences.

The cosine similarity between two embeddings AAA and BBB is calculated as:

$$\text{Similarity} = \frac{A \cdot B}{|A| \cdot |B|}$$

Where:

- $A \cdot B$ is the dot product of the two vectors.
- $|A|$ AND $|B|$ are the magnitudes (norms) of the vectors.

Interpretation of values:

- **1** → Backgrounds are highly similar (strong alignment of features)
- **0** → Backgrounds are unrelated (orthogonal features)
- **-1** → Backgrounds are opposite (features point in completely different directions)

A **similarity matrix of size $8,727 \times 8,727$** was computed, where each element (i, j) represents the similarity between the i^{th} and j^{th} background. This matrix forms the foundation for identifying visually dissimilar backgrounds.

3.3 Identification of Dissimilar Backgrounds

Using the similarity matrix, backgrounds with **low similarity scores** were selected for replacement experiments. These dissimilar backgrounds provide meaningful contrast when composited with foreground objects, enabling a controlled study of how background context influences model predictions.

Additionally, the similarity matrix allows for **quantitative analysis** of vision models: by comparing semantic predictions on original and replaced backgrounds, we can measure the degree to which models rely on background information.

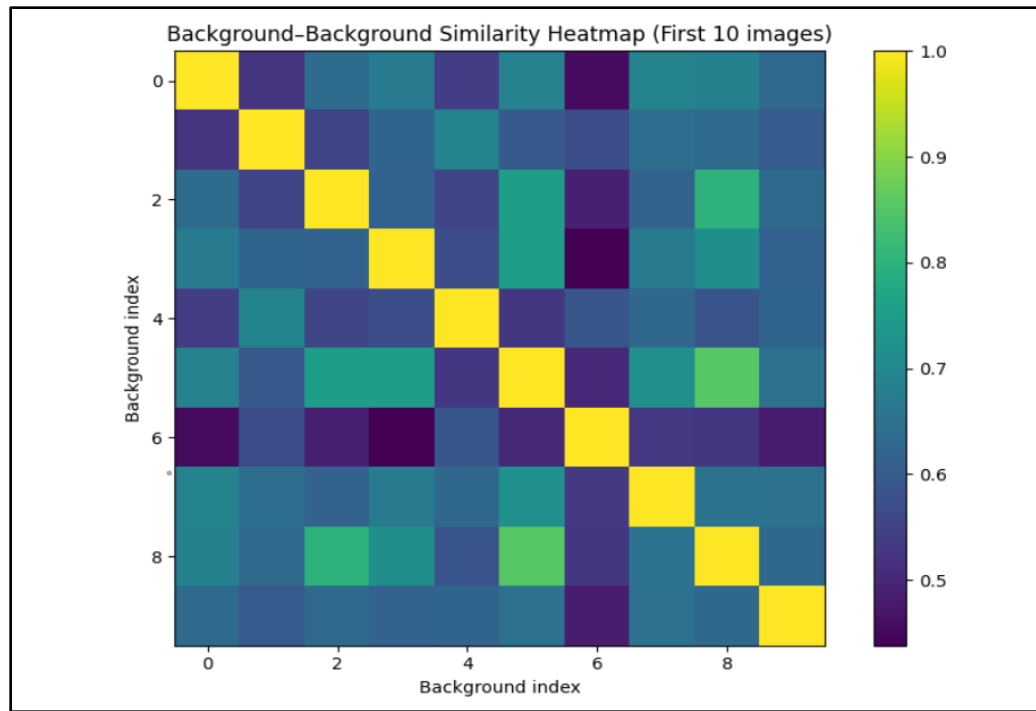


Figure 3.1 : Illustration of similarity matrix showing clusters of visually similar backgrounds and isolated dissimilar samples.

4. Background Replacement and Semantic Evaluation (Week 3)

In this phase, we investigated how replacing image backgrounds affects vision model predictions. The workflow involved selecting visually dissimilar backgrounds, compositing foreground objects onto these backgrounds, and evaluating the semantic output using the **CLIP** vision-language model.

4.1 Background Replacement Pipeline

For each foreground object, a **new background** was selected from the set of visually dissimilar images identified in Section 3. To ensure meaningful contrast, only backgrounds with **low cosine similarity scores** relative to the original were chosen. This selection strategy guarantees that the replacement background is significantly different in terms of texture, color, and scene context.

The replacement pipeline follows these steps:

1. Identify dissimilar backgrounds based on similarity threshold.
2. Pair each foreground object with a selected background.
3. Prepare foreground and background images for compositing.

4.2 Image Compositing Techniques

The compositing process integrates the foreground object with the new background using **alpha blending**, guided by the binary masks generated in Section 2.

Processing Steps

1. **Foreground Resizing:** The foreground object was resized proportionally to fit the dimensions of the new background.
2. **Mask Cleaning:** Binary masks were refined using morphological operations to remove noise and ensure smooth edges.
3. **Alpha Blending:** Foreground pixels were blended with the background based on the alpha mask, producing a single cohesive image.

Alpha blending preserves foreground transparency while merging seamlessly with the background, allowing visual continuity between the object and new environment.

4.3 Limitations of Compositing

While alpha blending enables automated foreground-background merging, several **visual artifacts** can reduce realism:

- **Lighting Mismatch:** Differences in brightness and shadows between foreground and new background.
- **Edge Artifacts:** Slight haloing or unnatural edges along the foreground boundary.
- **Scale Inconsistencies:** Objects may appear unnaturally large or small relative to the new scene.
- **Depth Misalignment:** Foreground objects do not always align with the perceived depth of the background.

These limitations highlight challenges in fully automated background replacement and emphasize that visual realism is constrained by available image preprocessing techniques.

4.4 Semantic Evaluation using CLIP (ViT-B/32)

To measure how background replacement influences model perception, **CLIP (ViT-B/32)** was employed. CLIP is a **vision-language model** trained on millions of image-text pairs, capable of computing the semantic similarity between an image and textual labels.

Each modified image was evaluated to determine whether the semantic content remained consistent or shifted due to background changes. CLIP outputs **probability scores** for a predefined set of labels, which allows quantitative analysis of model robustness to contextual changes.

4.5 Label Set and Confidence Analysis

A representative set of labels was selected to cover both **object categories** and **scene types**:

- **Objects:** person, car, bed, animal, vehicle
- **Scenes/Context:** room, indoor, outdoor

For each image, CLIP produced probability scores for these labels. These scores were analyzed to:

1. Detect cases where background changes caused significant shifts in predictions.
2. Evaluate the **confidence** of the model in identifying the foreground object regardless of background context.
3. Identify potential **context-based biases** in model predictions, highlighting the reliance on environmental cues.

This analysis provides a systematic method for assessing how much semantic understanding in vision models depends on background context versus foreground object features.

5. Results and Analysis

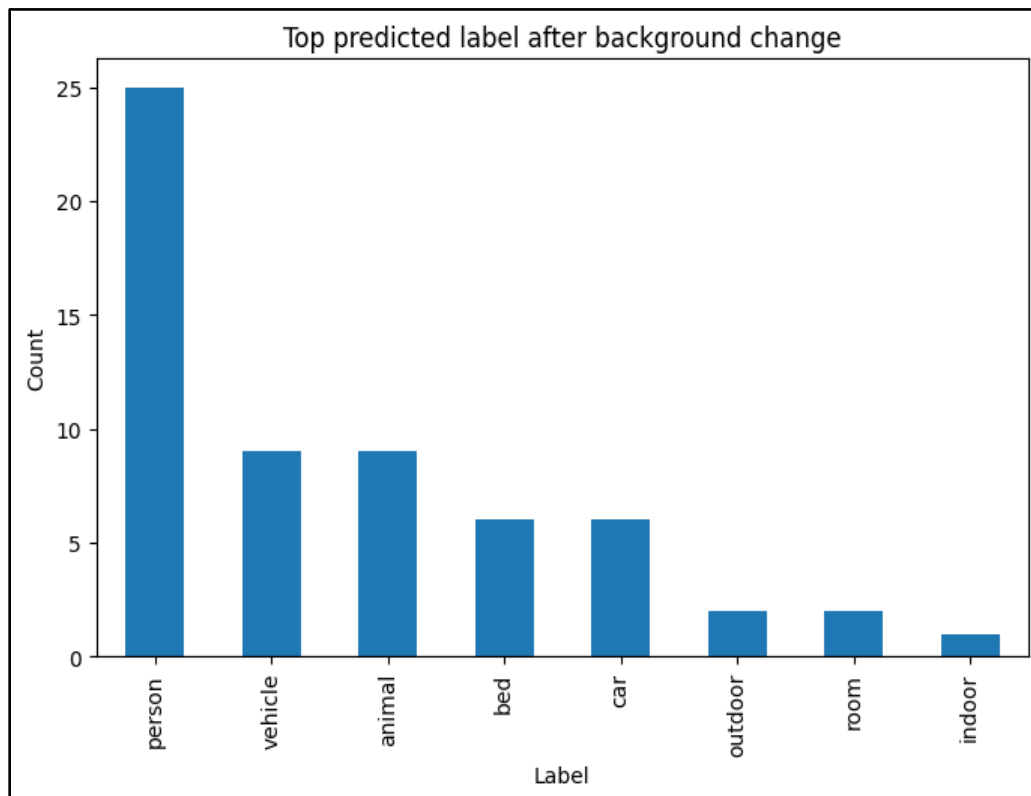
5.1 Dominant Label Distribution

For each modified image, the **dominant predicted label**—the label with the highest CLIP confidence score—was extracted.

Observations:

- The **person** class appeared most frequently as the dominant prediction, indicating that human foreground objects strongly influence semantic interpretation.
- **Vehicle** and **animal** classes were next most common, reflecting object-centric scenes in the dataset.
- **Indoor** scenes appeared least frequently, suggesting that environmental context alone has lower predictive dominance compared to foreground objects.

This analysis suggests that strong foreground objects can dominate model predictions even when the background is significantly altered.



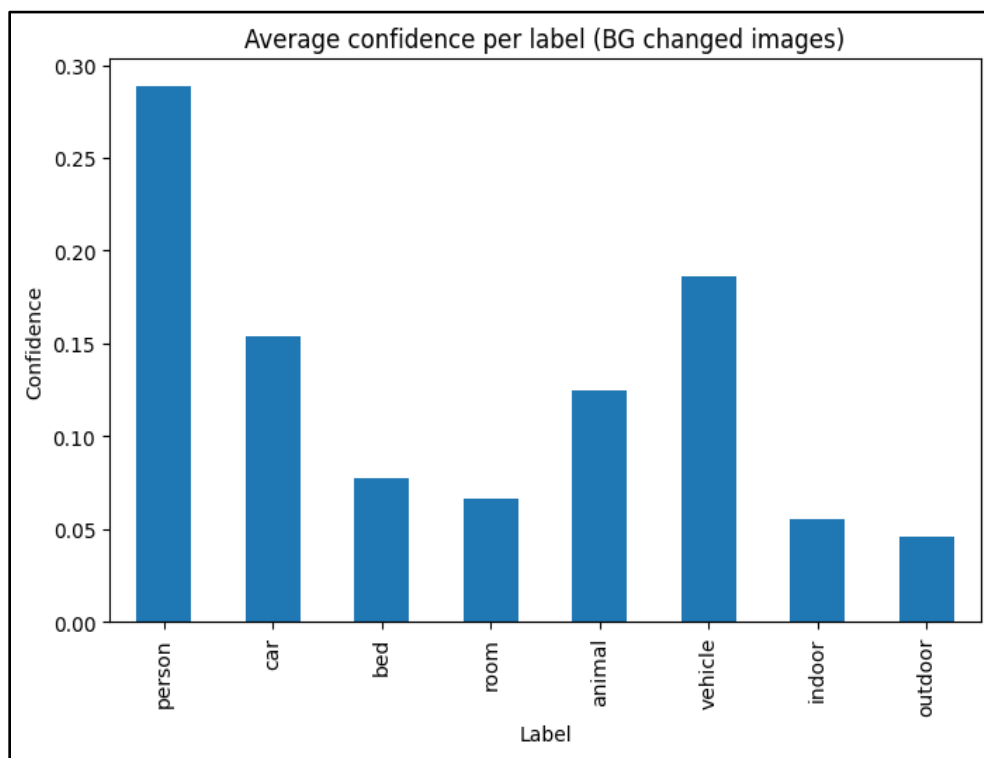
5.2 Average Confidence per Label

The **mean confidence score** for each label was computed across all images to evaluate the model's certainty in its predictions.

Observations:

- The **person** class achieved the highest average confidence, consistent with the dataset's human-centered imagery.
- Environment-related labels such as **outdoor** had lower average confidence, highlighting that background context alone provides weaker semantic cues.

High confidence scores for object-centric labels indicate that foreground objects play a primary role in the model's semantic understanding.



5.3 Background Sensitivity Analysis

To quantify the effect of background changes, a **confidence gap metric** was defined:

$$\text{Confidence Gap} = \text{Max}(\text{label score}) - \text{Min}(\text{label score})$$

- A higher confidence gap indicates stronger shifts in semantic predictions due to background replacement.

Observations:

- The largest confidence gaps occurred in scenes containing **cars**, **animals**, and **beds**, indicating that these object-centric images are more sensitive to background context.
- Foreground objects such as **persons** showed lower sensitivity, consistent with their dominance in semantic interpretation.

This analysis provides quantitative evidence of context-dependent behavior in vision models.

6. Limitations

Several factors influenced the experimental outcomes:

- **Imperfect Foreground Segmentation:** Residual background in masks may bias results.
- **Compositing Artifacts:** Edge halos and blending inconsistencies reduce realism.
- **Lighting and Color Mismatch:** Foreground and new backgrounds often had differing lighting conditions.
- **Lack of Depth-Aware Compositing:** Foreground objects may appear misaligned with scene depth.
- **Limited Label Set:** Semantic evaluation was restricted to a small set of labels.

These limitations may have affected the reliability of CLIP predictions and the realism of synthesized images.

7. Key Learnings

The project yielded several important insights into vision model behavior:

1. **Background Context Matters:** Semantic predictions can be influenced by background changes, especially for object-centric scenes.
2. **Foreground Dominance:** Humans and other prominent objects strongly drive model outputs, often overpowering background influence.
3. **Compositing Quality Affects Evaluation:** Naive blending introduces artifacts that may distort model predictions.
4. **Importance of Realism:** More realistic image synthesis improves the validity of experiments assessing contextual bias.

8. Conclusion

This study explored the impact of background context on vision model predictions using a large-scale dataset and a structured pipeline for foreground extraction, similarity analysis, and semantic evaluation.

- Foreground extraction and similarity-based background replacement were effective for systematic experiments.
- The experiments demonstrate that vision models rely on both foreground objects and contextual background cues.
- Limitations in compositing realism constrain the strength of conclusions, but the results highlight the **importance of considering contextual bias** in model design and evaluation.

Overall, background context can influence semantic interpretation, emphasizing the need for careful dataset and model analysis in computer vision.

9. Future Work

Potential improvements for extending this work include:

- **Depth-Aware Compositing:** Aligning foreground objects with background depth for more realistic scenes.
- **Poisson or Gradient-Domain Blending:** Seamless integration of foreground and background.
- **Diffusion-Based Background Generation:** Generating realistic and varied backgrounds automatically.
- **Cross-Architecture Testing:** Evaluating contextual bias across multiple vision models beyond CLIP.
- **Expanded Label Set:** Including more object and scene categories for richer semantic evaluation.

References

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